

SZOMBATHELY AP VAS, Hungary

WMO: 128120

Lat: 47.1983N

Lon: 16.6478E

Elev: 221

StdP: 98.70

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.6	-8.1	-14.7	1.1	-9.2	-11.9	1.4	-6.5	14.2	-0.7	12.1	-0.2	2.5	10	0.501

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	32.2	20.8	30.4	20.5	28.7	19.8	22.1	29.1	21.2	28.4	20.4	27.2	3.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.8	14.9	24.9	18.9	14.1	23.9	18.1	13.4	23.1	65.6	29.0	62.5	28.4	59.6	27.3	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.2	6.9	-14.4	34.8	3.4	2.0	-16.8	36.3	-18.8	37.5	-20.6	38.6	-23.1	40.0		
(5)			-14.7	23.7	3.2	1.1	-17.0	24.5	-18.9	25.2	-20.7	25.8	-23.1	26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-0.2	1.7	5.9	10.9	15.3	19.1	21.0	20.5	15.8	10.8	5.8	0.8		
	DBStd	8.34	4.15	4.52	4.35	3.77	3.23	3.39	3.06	3.07	3.19	3.74	4.06	3.97		
	HDD10.0	1181	317	233	137	34	2	0	0	0	1	37	135	287		
	HDD18.3	3050	575	466	386	223	102	31	12	13	87	234	377	545		
	CDD10.0	1424	0	1	9	61	167	273	340	327	176	61	8	1		
	CDD18.3	252	0	0	0	1	9	55	93	81	13	1	0	0		
	CDH23.3	2645	0	0	0	17	138	566	958	833	127	7	0	0		
	CDH26.7	876	0	0	0	1	21	169	348	315	22	0	0	0		
(14) Wind	WSAvg	2.8	2.6	3.0	3.2	3.4	3.0	2.8	2.7	2.4	2.5	2.3	2.6	2.7		
(15) Precipitation	PrecAvg	650	27	28	35	45	67	81	78	75	66	57	53	38		
	PrecMax	888	65	89	75	85	125	158	146	161	156	133	132	77		
	PrecMin	472	8	5	2	1	22	21	6	11	20	2	5	6		
	PrecStd	101	16	24	21	20	28	32	39	39	38	34	28	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	16.4	20.4	25.6	28.8	32.7	34.7	34.9	29.1	24.2	18.6	13.1		
		MCWB	8.1	9.6	11.3	16.0	18.4	20.9	21.5	20.6	19.5	16.8	13.7	10.0		
	2%	DB	9.4	13.1	18.0	22.8	26.6	30.6	32.1	32.4	26.5	21.5	15.6	9.5		
		MCWB	6.5	8.2	10.8	14.2	18.0	20.9	20.9	20.7	18.3	16.0	12.1	7.3		
	5%	DB	7.2	10.6	15.6	20.7	24.8	28.7	30.3	30.2	24.6	19.4	13.4	7.5		
		MCWB	4.9	6.9	9.8	13.2	17.1	20.1	20.4	20.5	17.8	14.8	11.0	5.5		
	10%	DB	5.2	8.3	13.2	18.8	22.7	26.8	28.5	28.0	22.5	17.2	11.5	5.8		
		MCWB	3.7	5.3	8.4	12.0	15.8	19.1	19.7	19.5	16.5	13.7	9.5	4.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	10.1	12.5	16.6	20.1	23.0	23.2	23.2	20.6	17.8	14.3	10.2		
		MCDB	11.8	15.5	18.3	24.2	26.6	29.7	30.7	29.8	27.2	22.8	17.8	12.6		
	2%	WB	6.8	8.5	11.2	14.9	18.7	21.7	21.9	22.0	19.4	16.4	12.5	7.6		
		MCDB	9.0	12.5	16.7	21.5	25.0	28.3	29.4	29.0	24.9	20.5	15.1	9.2		
	5%	WB	5.2	7.1	10.1	13.7	17.6	20.6	21.1	21.1	18.4	15.2	11.2	5.8		
		MCDB	6.8	10.2	14.9	19.3	23.4	27.2	28.4	28.4	23.0	18.7	13.3	6.9		
	10%	WB	3.9	5.6	8.9	12.6	16.5	19.7	20.3	20.1	17.2	14.1	9.8	4.7		
		MCDB	5.0	8.0	12.7	17.9	21.7	25.7	27.1	26.9	21.4	16.8	11.2	5.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	7.9	9.8	11.2	11.0	11.4	11.8	11.8	10.3	9.8	6.6	5.5		
		MCDBR	9.2	12.3	14.4	15.6	15.0	14.7	15.4	15.8	13.7	12.9	10.3	8.3		
	5% WB	MCWBR	6.5	8.0	8.4	8.1	7.4	6.2	5.6	5.8	6.6	7.6	7.1	6.2		
		MCWBR	8.7	11.4	13.2	13.8	13.5	13.6	13.8	14.2	12.2	12.0	9.7	7.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.320	0.344	0.381	0.426	0.422	0.427	0.431	0.434	0.409	0.380	0.350	0.320		
		taud	2.390	2.343	2.268	2.160	2.222	2.248	2.237	2.241	2.283	2.338	2.375	2.414		
	Ebn at Noon	Ebn at Noon	758	813	833	826	844	841	830	810	794	758	709	712		
		Edh at Noon	71	91	114	139	136	133	133	128	112	91	72	63		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.11	1.92	3.13	4.54	5.44	5.92	5.82	5.16	3.67	2.35	1.25	0.85		
	RadStd	RadStd	0.12	0.27	0.38	0.44	0.52	0.41	0.44	0.51	0.43	0.24	0.13	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (85 neighbors)	+0.57	N/A	N/A	+0.90	+0.49	+0.44	-116	-160	+97	+57

Nomenclature: See separate page